

PRACTICAL : 9

AIM : Cleaning and Structuring Data for Analysis

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

Dataset: Custom Designed Food_Delivery Dataset [csv files]

Food_Delivery_Orders.csv

Order_ID, Customer_Name, Restaurant_Code, Order_Value, Delivery_Time, Food_Rating, Delivery_Rating, Discount, Customer_Satisfaction

O101,Aarav Sharma,RES-01,450,32,4.5,4.2,10%,85 %
O102,Priya Verma,RES-02,,28,4.0,4.5,15%,90 %
O103,Rohan Gupta,RES-01,620,45,,3.8,5%,78 %
O104,Sneha Patel,RES-03,380,25,4.8,4.7,20%,95 %
O105,Aditya Rao,RES-04,750,,3.5,3.2,25%,65 %
O106,Ananya Singh,RES-01,520,35,4.6,4.4,10%,88 %
O107,Rahul Nair,RES-02,,40,3.8,3.9,15%,72 %
O108,Kavya Reddy,RES-03,890,22,4.9,4.8,30%,98 %
O109,Vikram Joshi,RES-01,340,50,3.2,3.0,5%,80 %
O110,Pooja Iyer,RES-02,580,30,4.3,4.1,20%,91 %s

Food_Delivery_Orders_2.csv

Order_ID, Customer_Name, Restaurant_Code, Order_Value, Delivery_Time, Food_Rating, Delivery_Rating, Discount, Customer_Satisfaction

O201,Aarav Sharma,RES-01,520,30,4.6,4.4,12%,88 %
O202,Priya Verma,RES-02,680,35,4.2,4.3,18%,92 %
O203,Rohan Gupta,RES-01,590,40,4.3,4.0,10%,82 %
O204,Sneha Patel,RES-03,420,24,4.9,4.8,25%,96 %

SHETH L.U.J. AND SIR M.V. COLLEGE

O205,Aditya Rao,RES-04,810,38,3.8,3.7,30%,70 %
O206,Ananya Singh,RES-01,560,32,4.7,4.5,15%,90 %
O207,Rahul Nair,RES-02,490,42,4.0,4.1,20%,75 %
O208,Kavya Reddy,RES-03,920,20,5.0,4.9,35%,99 %
O209,Vikram Joshi,RES-01,390,48,3.5,3.4,8%,83 %
O210,Pooja Iyer,RES-02,610,29,4.4,4.2,22%,94 %

Restaurant_Master.csv

Restaurant_Code,Restaurant_Name,Cuisine,City,Target_Rating,Daily_Order_Capacity
RES-01,Spice Garden,Indian,Mumbai,4.3,150
RES-02,Urban Tadka,North Indian,Pune,4.1,120
RES-03,Food Street,Multi-Cuisine,Mumbai,4.5,100
RES-04,The Food Hub,Continental,Delhi,4.0,80

Customer_Activity.csv

Customer_Name,Loyalty_Program,Orders_Completed,Offers_Used,App_Rating
Aarav Sharma,Gold,12,5,4.6
Priya Verma,Silver,8,4,4.3
Rohan Gupta,Gold,10,3,4.1
Sneha Patel,Platinum,15,7,4.8
Aditya Rao,Bronze,5,2,3.7
Ananya Singh,Gold,13,5,4.7
Rahul Nair,Silver,7,3,4.0
Kavya Reddy,Platinum,18,8,4.9
Vikram Joshi,Bronze,6,2,3.6
Pooja Iyer,Gold,11,5,4.5

A: Identify and Handle Missing or Inconsistent Data

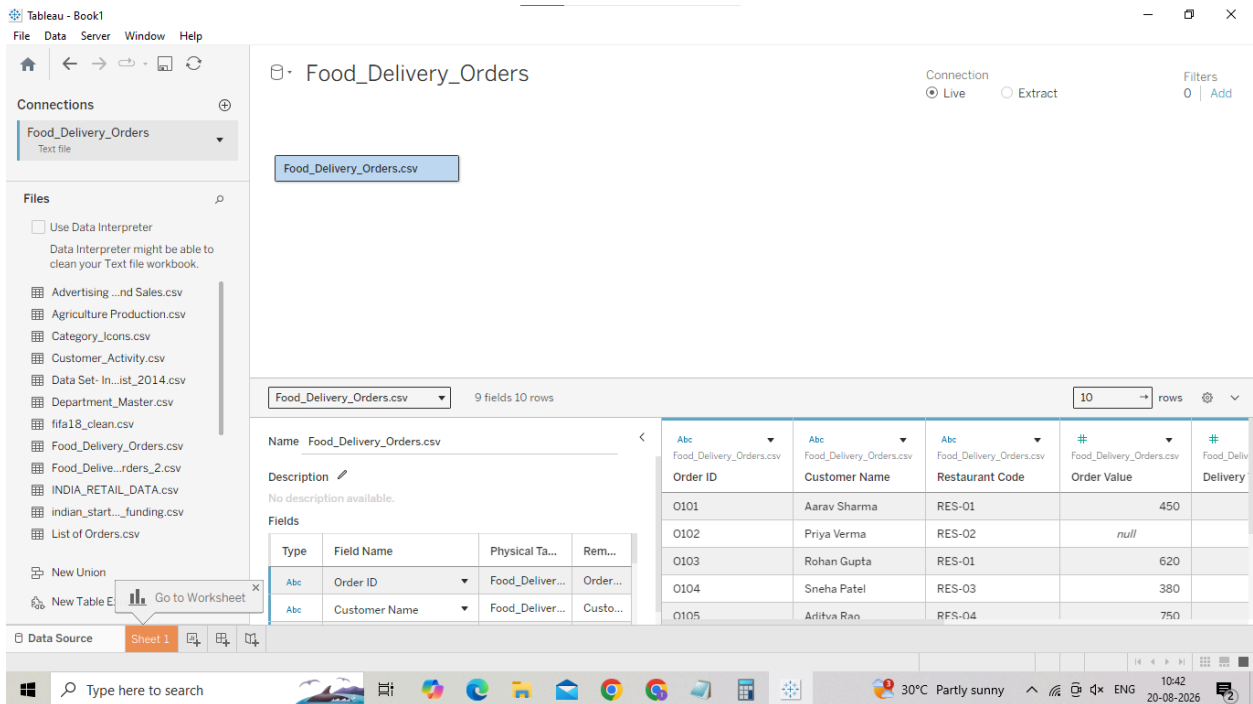


Tableau - Book1

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising ...nd Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

New Union

New Table E

Go to Worksheet

Data Source

Sheet 1

Food_Delivery_Orders

Food_Delivery_Orders.csv

9 fields 10 rows

10 rows

Name Food_Delivery_Orders.csv

Description

No description available.

Fields

Type	Field Name	Physical Ta...	Rem...
Abc	Order ID	Food_Deliver...	Order...
Abc	Customer Name	Food_Deliver...	Custo...

Order ID	Customer Name	Restaurant Code	Order Value	Food_Deliv...
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Aaritva Rao	RFS-04	750	

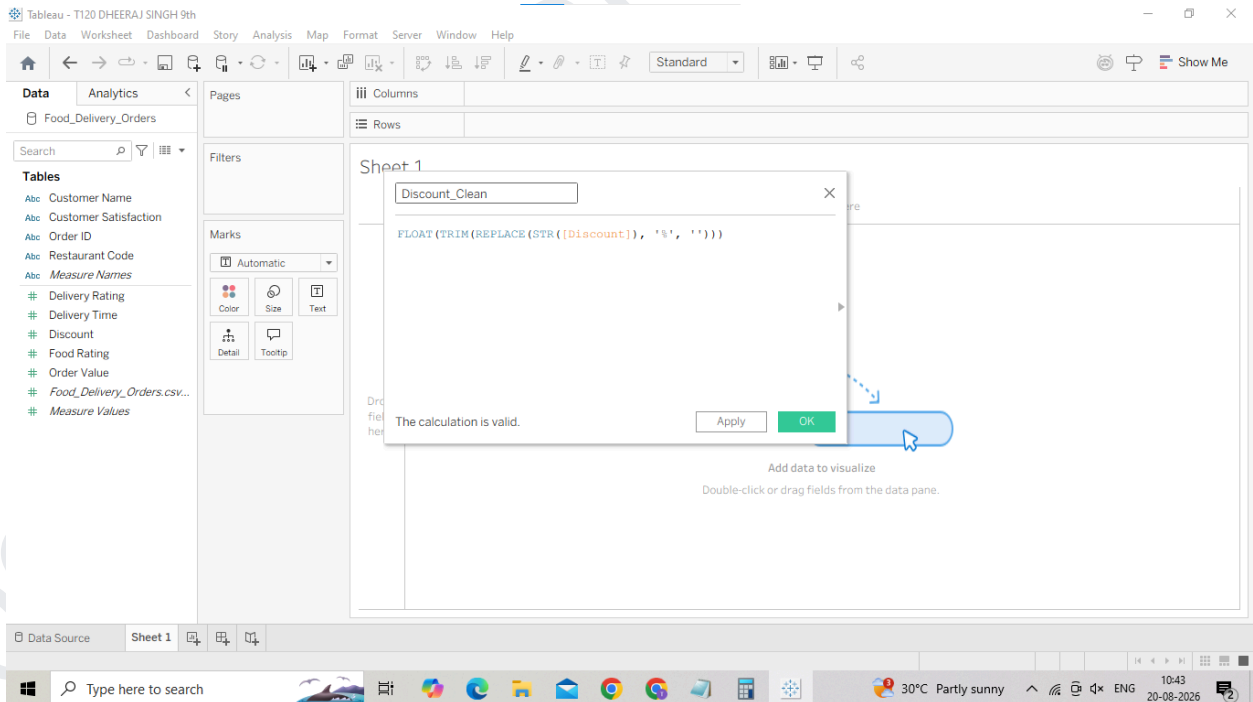


Tableau - T120 DHEERAJ SINGH 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Data Analytics

Food_Delivery_Orders

Search

Tables

- Abc Customer Name
- Abc Customer Satisfaction
- Abc Order ID
- Abc Restaurant Code
- Abc Measure Names
- # Delivery Rating
- # Delivery Time
- # Discount
- # Food Rating
- # Order Value
- # Food_Delivery_Orders.csv...
- # Measure Values

Columns

Rows

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Discount_Clean

Float (Trim (Replace (Str ([Discount]), '%', '')))

The calculation is valid.

Apply OK

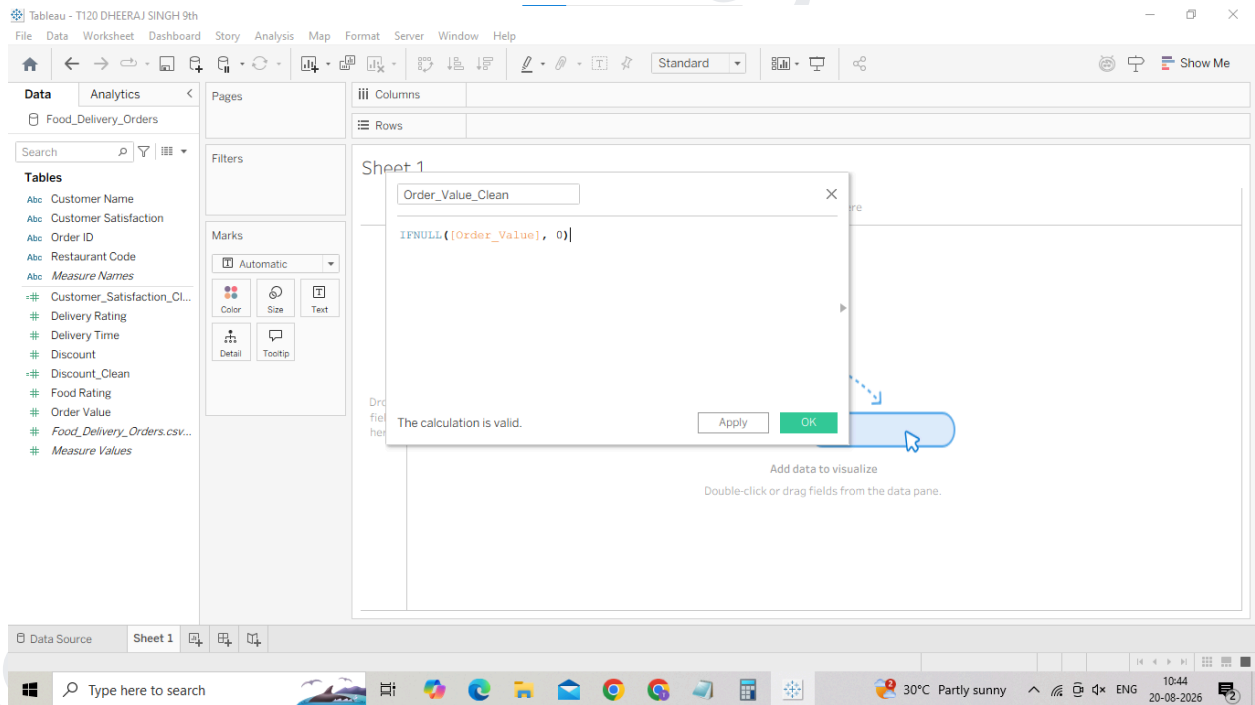
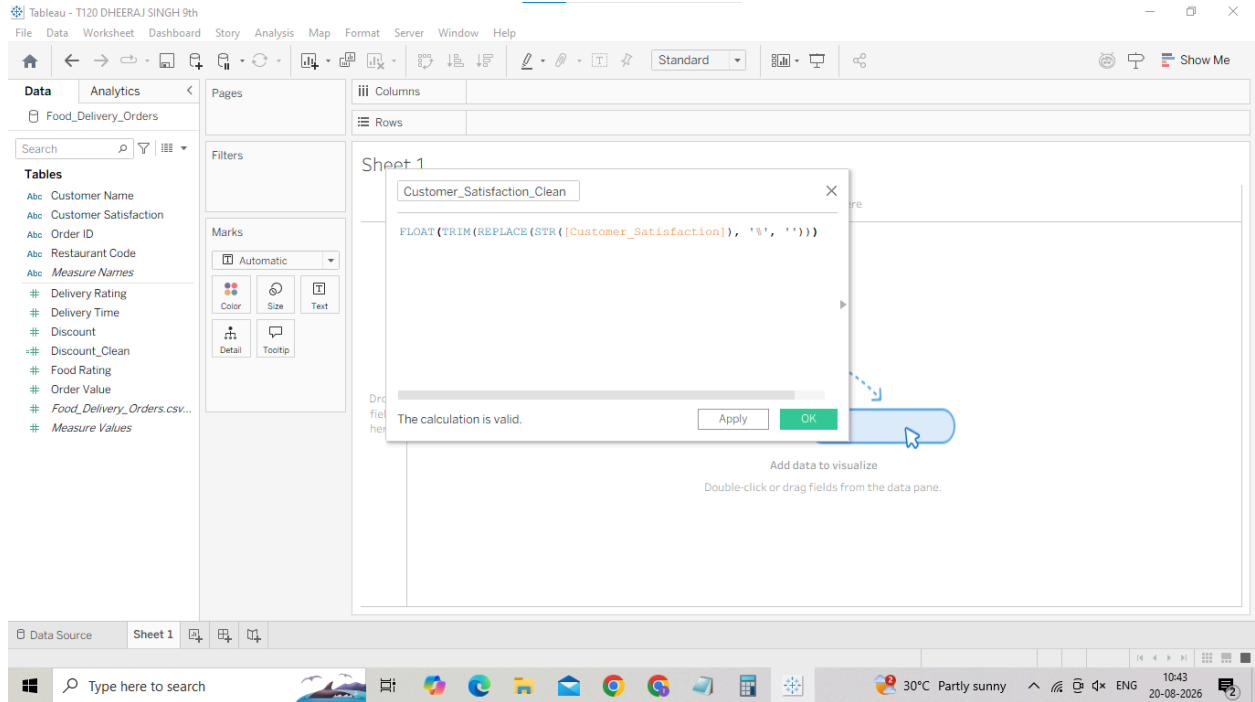
Add data to visualize

Double-click or drag fields from the data pane.

Data Source

Sheet 1

SHETH L.U.J. AND SIR M.V. COLLEGE



SHETH L.U.J. AND SIR M.V. COLLEGE

This screenshot shows the Tableau Desktop interface. The 'Data' pane on the left lists fields from the 'Food_Delivery_Orders' data source, including 'Customer Name', 'Customer Satisfaction', 'Order ID', 'Restaurant Code', 'Measure Names', 'Customer_Satisfaction_Cli...', 'Delivery Rating', 'Delivery Time', 'Discount', 'Discount_Clean', 'Food Rating', 'Food_Rating_Clean', 'Order Value', 'Order_Value_Clean', 'Food_Delivery_Orders.csv...', and 'Measure Values'. The 'Marks' card is set to 'Automatic'. A dialog box titled 'Food_Rating_Clean' is open, displaying the formula `IFNULL([Food_Rating], 0)`. The dialog box includes a 'The calculation is valid.' message and 'Apply' and 'OK' buttons. The background shows a blank worksheet with 'Columns' and 'Rows' shelves.

This screenshot shows the Tableau Desktop interface. The 'Data' pane on the left lists fields from the 'Food_Delivery_Orders' data source, including 'Customer Name', 'Customer Satisfaction', 'Order ID', 'Restaurant Code', 'Measure Names', 'Customer_Satisfaction_Cli...', 'Delivery Rating', 'Delivery Time', 'Discount', 'Discount_Clean', 'Food Rating', 'Food_Rating_Clean', 'Order Value', 'Order_Value_Clean', 'Food_Delivery_Orders.csv...', and 'Measure Values'. The 'Marks' card is set to 'Automatic'. A dialog box titled 'Delivery_Time_Clean' is open, displaying the formula `IFNULL([Delivery_Time], 0)`. The dialog box includes a 'The calculation is valid.' message and 'Apply' and 'OK' buttons. The background shows a blank worksheet with 'Columns' and 'Rows' shelves.

B: Convert Wide Data into Tall Format

Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising...nd Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
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- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Food_Delivery_Orders.csv 14 fields 10 rows

10 rows

Name Food_Delivery_Orders.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Order ID	Food_Delivery_...	Order...
Abc	Customer Name	Food_Delivery_...	Custo...

Order ID	Customer Name	Restaurant Code	Order Value	Deliver
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Aarav Rao	RES-04	750	

Data Source Sheet 1

Type here to search

They found cold resc...

10:49
20-08-2026

Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

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New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Relate tables
Drag tables near Food_Delivery_Orders.csv to relate them.

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Food_Delivery_Orders.csv 14 fields 10 rows

10 rows

Name Food_Delivery_Orders.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Restaurant Code	Food_Delivery_...	Resto...
Abc	Order Value	Food_Delivery_...	Order...
Abc	Delivery Time	Food_Delivery_...	Delive...
Abc	Food Rating	Food_Delivery_...	Food...
Abc	Delivery Rating	Food_Delivery_...	Delive...

Order ID	Customer Name	Restaurant Code	Order Value	Delivery Time	Food Rating	Delivery Rating	Discount
O101	Aarav Sharma	RES-01	450	32	4.50000	4.20000	0.1
O102	Priya Verma	RES-02	null	28	4.00000	4.50000	0.1
O103	Rohan Gupta	RES-01	620	45	null	3.80000	0.0
O104	Sneha Patel	RES-03	380	25	4.80000	4.70000	0.2
O105	Aarav Rao	RES-04	750	null	3.50000	3.20000	0.1
O106	Ananya Singh	RES-01	520	35	4.60000	4.40000	0.1
O107	Rahul Nair	RES-02	null	40	3.80000	3.90000	0.1
O108	Kanya Reddy	RES-03	890	22	4.90000	4.80000	0.2

Data Source Sheet 1

Type here to search

10:50
20-08-2026

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Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

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- indian_start...funding.csv
- List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Food_Delivery_Orders.csv 12 fields 40 rows 40 rows

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Metric	Pivot	Pivot...
#	Value	Pivot	Pivot...
+ #	Discount_Clean	Calculation	Calcul...

Order ID	Customer Name	Restaurant Code	Discount	Customer Satisfaction
O101	Aarav Sharma	RES-01	0.100000	85 %
O101	Aarav Sharma	RES-01	0.100000	85 %
O101	Aarav Sharma	RES-01	0.100000	85 %
O101	Aarav Sharma	RES-01	0.100000	85 %
O102	Priya Verma	RES-02	0.150000	90 %
O102	Priva Verma	RES-02	0.150000	90 %

Data Source Sheet 1

Type here to search

30°C Partly sunny 10:51 20-08-2026

Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising ...nd Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Food_Delivery_Orders.csv 12 fields 40 rows 40 rows

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Metric	Pivot	Pivot...
#	Value	Pivot	Pivot...
+ #	Discount_Clean	Calculation	Calcul...

Customer Satisfaction	Metric	Pivot Value	Calculation	Calculation
00 85 %	Delivery Rating	4.200	0.100000	
00 85 %	Delivery Time	32.000	0.100000	
00 85 %	Food Rating	4.500	0.100000	
00 85 %	Order Value	450.000	0.100000	
00 90 %	Delivery Rating	4.500	0.150000	
00 90 %	Delivery Time	28.000	0.150000	

Data Source Sheet 1

Type here to search

30°C Partly sunny 10:52 20-08-2026

C: Perform Union of Multiple Files

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising ...nd Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Patient_Records.csv

Open...
Rename
Suggestions...
Remove
☒ Field names are in first row
Generate field names automatically
Text File Properties...
Convert to Union...

Patient_Records.csv 11 fields 10 rows

10 rows

Name Patient_Records.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Patient ID	Patient_Reco...	Patient...
Abc	Patient Name	Patient_Reco...	Patient...

Patient ID	Patient Name	Department	Age	Gender	Diagn...
P201	Rahul Mehta	Cardiology	54	Male	Hype...
P202	Anita Sharma	Orthopedics	42	Female	Fract...
P203	Vikram Singh	Neurology	61	Male	Migr...
P204	Priya Nair	Pediatrics	12	Female	Fever...
P205	Ariun Patel	Cardiolog...	67	Male	Hear...

Data Source Sheet 1

Type here to search

28°C Mostly cloudy

08:33 27-08-2026

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising ...nd Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

Patient_Records

Drag tables here to create a data model

[Learn more](#)

Data Source Sheet 1

Type here to search

28°C Mostly cloudy

08:35 27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delivery_Orders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table Extension

Patient_Records

Connection: Patient_Records

Food_Delivery_Orders.csv+
Specific (manual) Wildcard (automatic)

Food_Delivery_Orders.csv
Food_Delivery_Orders_2.csv

Tables in union: 2

Apply OK

Food_Delivery_Orders.csv

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Ta...	Rem...
Abc	Order ID	Food_Deliver...	Order...
Abc	Customer Name	Food_Deliver...	Custo...

20 rows

Abc	Abc	Abc	#	#
Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv
Order ID	Customer Name	Restaurant Code	Order Value	Delivery
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Arlitva Rao	RFS-04	750	

Data Source Sheet 1

Type here to search

28°C Mostly cloudy 08:37 27-08-2026

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delivery_Orders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table Extension

Patient_Records

Connection: Patient_Records

Food_Delivery_Or...

Food_Delivery_Orders.csv

10 fields 20 rows

20 rows

Abc	Abc	Abc	#	#
Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv
Order ID	Customer Name	Restaurant Code	Order Value	Delivery
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Arlitva Rao	RFS-04	750	

Data Source Sheet 1

Type here to search

28°C Mostly cloudy 08:37 27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - Dheeraj 9th

FileDataServerWindowHelp

Home

←

→

↶

↷

↺

Connections

Patient_Records

Text file

Files

☐ Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv

Agriculture Production.csv

Category_Icons.csv

Customer_Activity.csv

Data Set- In...ist_2014.csv

Department_Master.csv

fifa18_clean.csv

Food_Delivery_Orders.csv

Food_Delive...rders_2.csv

INDIA_RETAIL_DATA.csv

indian_start...funding.csv

List of Orders.csv

New Union

New Table Extension

Patient_Records

Connection

Live

Extract

Filters

0 | Add

Food_Delivery_Or...

Food_Delivery_Orders.csv

10 fields 20 rows

20 rows

Name Food_Delivery_Orders.csv

Description

No description available.

Fields

Type	Field Name	Physical Ta...	Rem...
Abc	Order ID	Food_Deliver...	Order...
Abc	Customer Name	Food_Deliver...	Custo...

#	#	Abc	Abc
Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv+	Food_Delivery_Orders.csv+
Delivery Rating	Discount	Customer Satisfaction	Table Name
4.20000	0.100000	85 %	Food_Delivery_Orders.csv
4.50000	0.150000	90 %	Food_Delivery_Orders.csv
3.80000	0.050000	78 %	Food_Delivery_Orders.csv
4.70000	0.200000	95 %	Food_Delivery_Orders.csv
3.20000	0.250000	65 %	Food Delivery Orders.csv

Data Source

Sheet 1

Type here to search

28°C Mostly cloudy

08:38 27-08-2026

D: Execute Cross-Database Joins

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

indian_start..._funding.csv
List of Orders.csv
Most Strea...olumns.csv
Orders.csv
Patient_Feedback.csv
Patient_Records.csv
Restaurant_Master.csv
startup_funding.csv
store_sales_50_rows.csv
Students - Students.csv
titanic.csv
Treatment_Records.csv

New Union
New Table Extension

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Deli... — Restaurant...

4 rows

Name Restaurant_Master.csv

How do relationships differ from joins? [Learn more](#)

Food_Delivery_Ord... Operator Restaurant_Master...

Abc Restaurant Cod... = Abc Restaurant Cod...

Add more fields

Performance Options

Restaurant Code (Restaur	Restaurant Name	Cuisine	City	Target
RES-01	Spice Garden	Indian	Mumbai	
RES-02	Urban Tadka	North Indian	Pune	
RES-03	Food Street	Multi-Cuisine	Mumbai	
RES-04	The Food Hub	Continental	Delhi	

Data Source Sheet 1

Type here to search

29°C Mostly cloudy

08:41
27-08-2026

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

indian_start..._funding.csv
List of Orders.csv
Most Strea...olumns.csv
Orders.csv
Patient_Feedback.csv
Patient_Records.csv
Restaurant_Master.csv
startup_funding.csv
store_sales_50_rows.csv
Students - Students.csv
titanic.csv
Treatment_Records.csv

New Union
New Table Extension

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Deli... — Restaurant...

4 rows

Name Restaurant_Master.csv

How do relationships differ from joins? [Learn more](#)

Food_Delivery_Ord... Operator Restaurant_Master...

Abc Restaurant Cod... = Abc Restaurant Cod...

Add more fields

Performance Options

Restaurant Code (Restaur	Restaurant Name	Cuisine	City	Target
RES-01	Spice Garden	Indian	Mumbai	
RES-02	Urban Tadka	North Indian	Pune	
RES-03	Food Street	Multi-Cuisine	Mumbai	
RES-04	The Food Hub	Continental	Delhi	

Data Source Sheet 1

Type here to search

29°C Mostly cloudy

08:41
27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

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- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv
- Most Strea...olumns.csv

New Union

New Table Extension

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Deli... — Customer...

10 rows

Name Customer_Activity.csv

How do relationships differ from joins? [Learn more](#)

Food_Delivery_Ord... Operator Customer_Activity...

Abc Customer Name ▼ = ▼ Abc Customer Name ▼

+ Add more fields

> Performance Options

Abc Customer_Activity.csv	Abc Customer_Activity.csv	# Customer_Activity.csv	# Customer_Activity.csv
Customer Name (Customi	Loyalty Program	Orders Completed	Offers Used
Aarav Sharma	Gold	12	5
Priya Verma	Silver	8	4
Rohan Gupta	Gold	10	3
Sneha Patel	Platinum	15	7
Aritiva Rao	Bronze	5	2

Data Source Sheet 1

Type here to search

29°C Mostly cloudy

08:42 27-08-2026

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv
- Most Strea...olumns.csv

New Union

New Table Extension

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Deli... — Customer...

10 rows

Name Customer_Activity.csv

How do relationships differ from joins? [Learn more](#)

Food_Delivery_Ord... Operator Customer_Activity...

Abc Customer Name ▼ = ▼ Abc Customer Name ▼

+ Add more fields

> Performance Options

Abc Customer_Activity.csv	Abc Customer_Activity.csv	# Customer_Activity.csv	# Customer_Activity.csv
Customer Name (Customi	Loyalty Program	Orders Completed	Offers Used
Aarav Sharma	Gold	12	5
Priya Verma	Silver	8	4
Rohan Gupta	Gold	10	3
Sneha Patel	Platinum	15	7
Aritiva Rao	Bronze	5	2

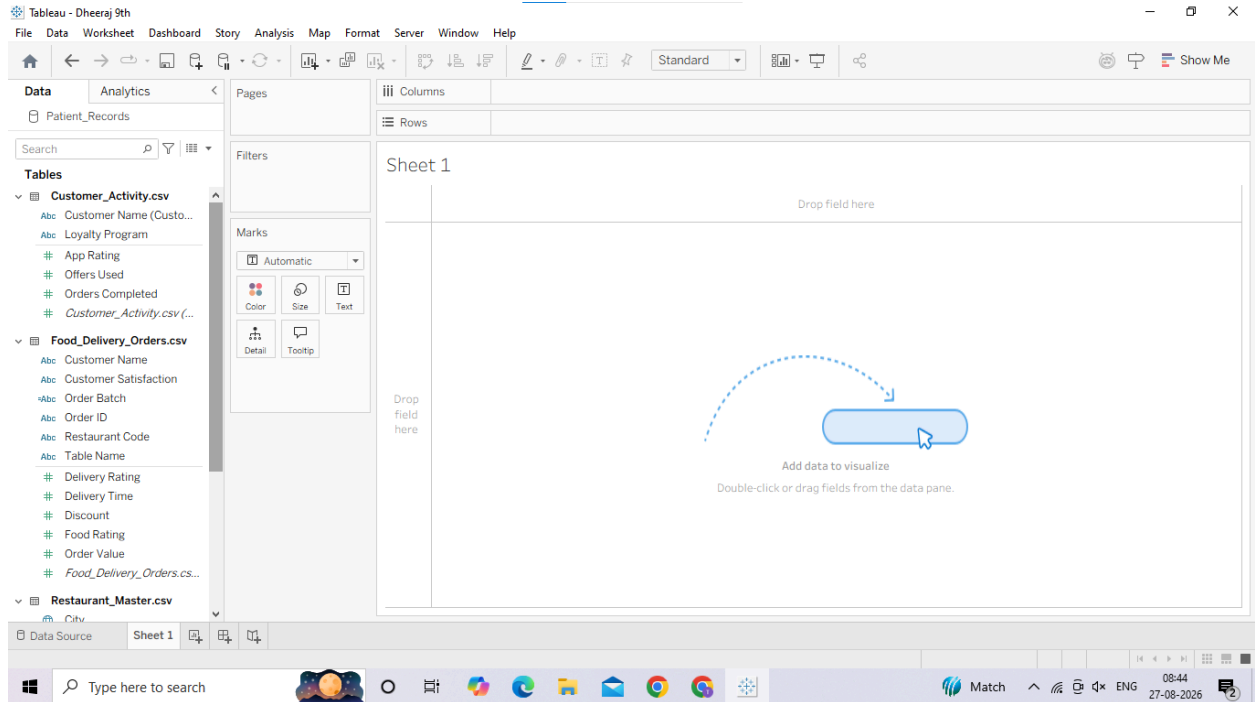
Data Source Sheet 1

Type here to search

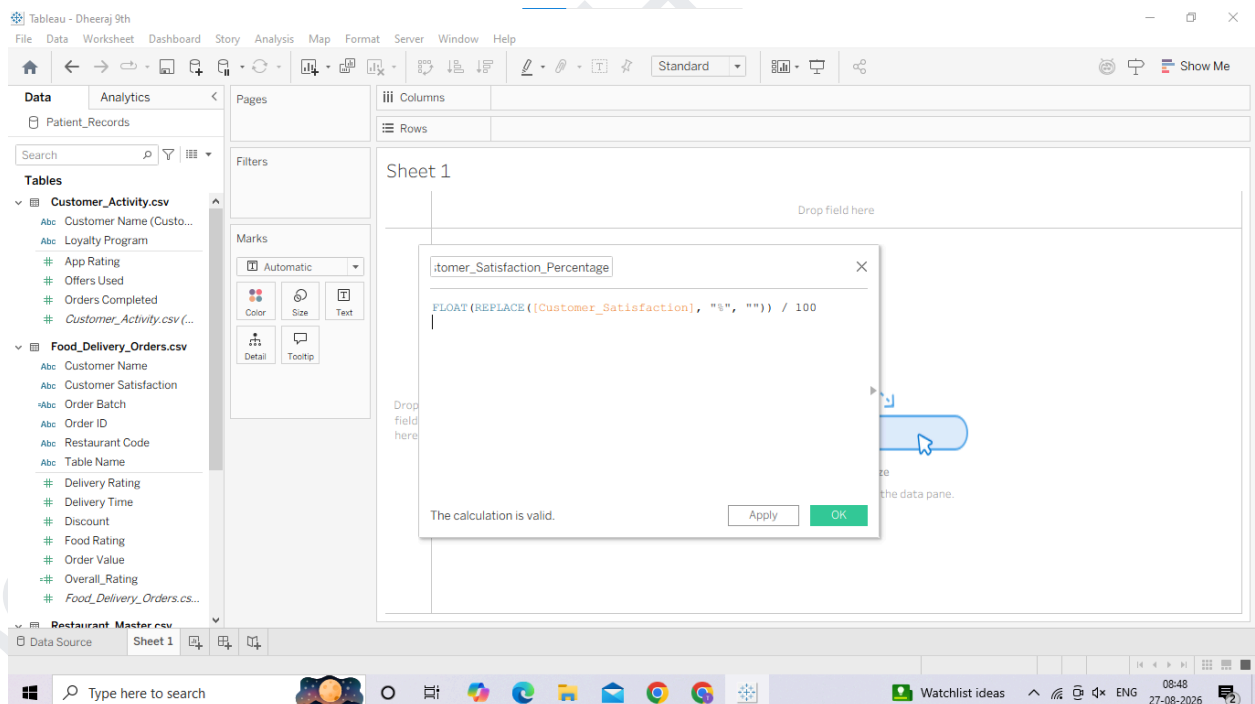
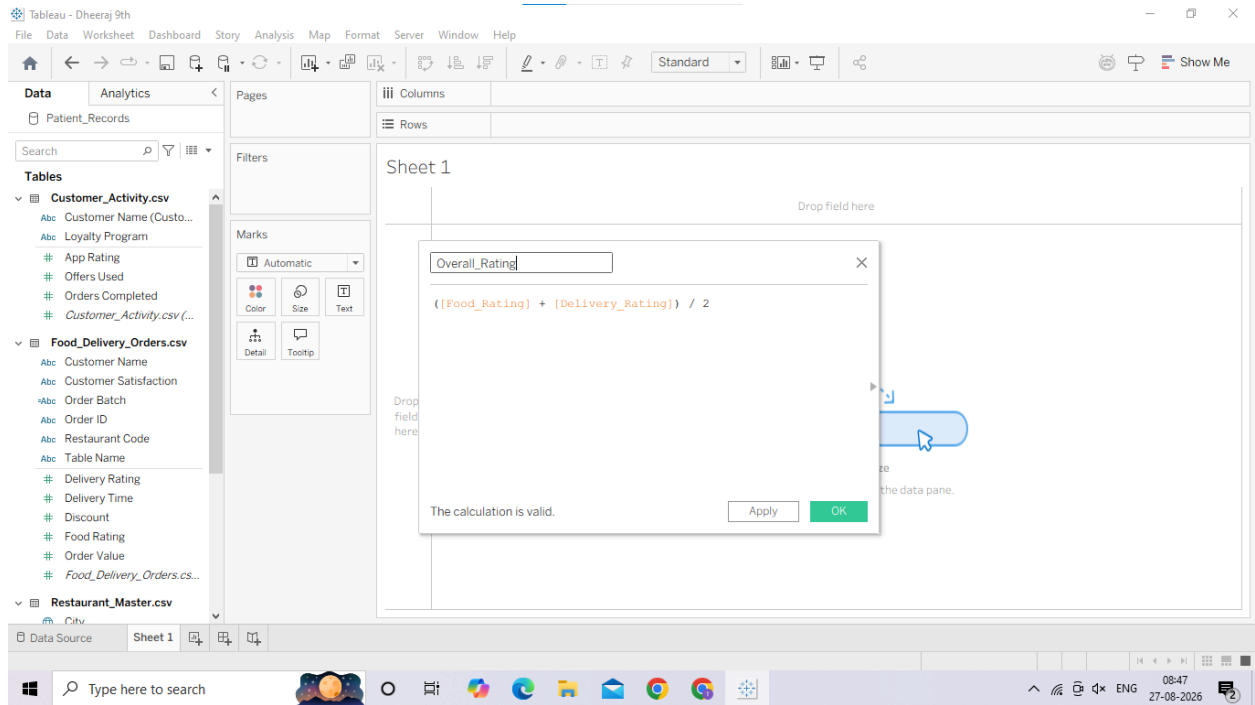
Gold +0.69%

08:43 27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE



E: Restructure Data for Better Visualization



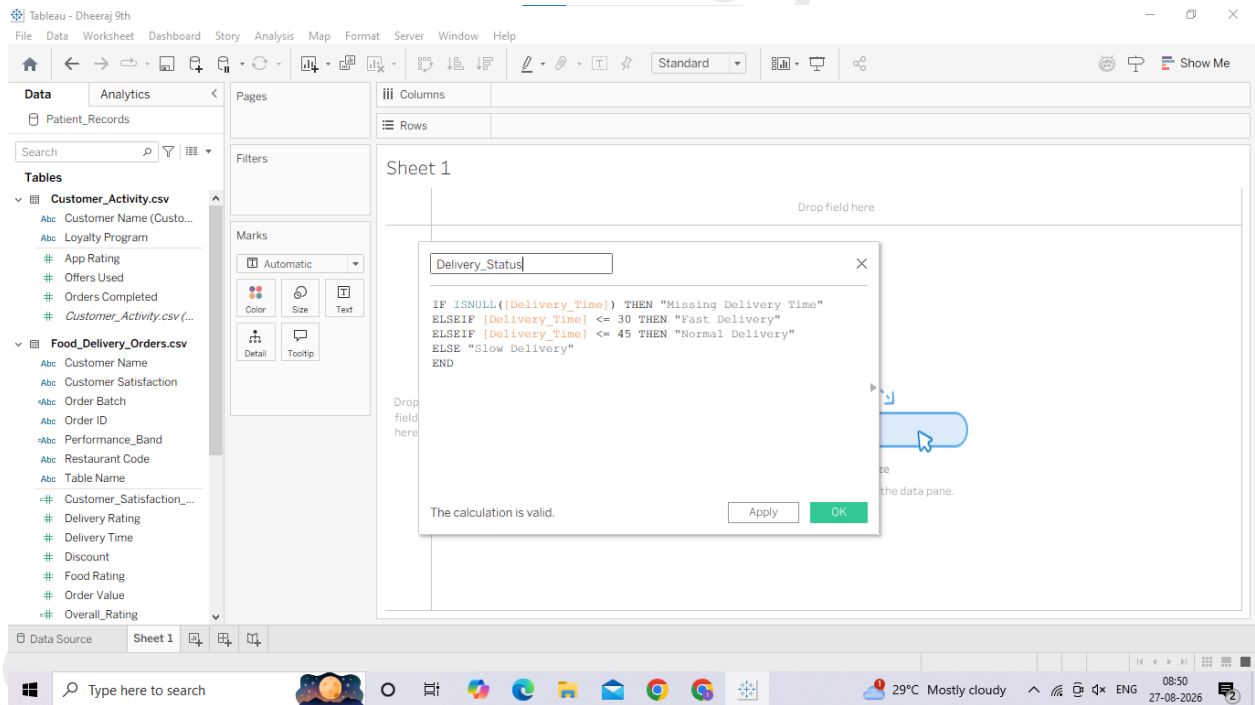
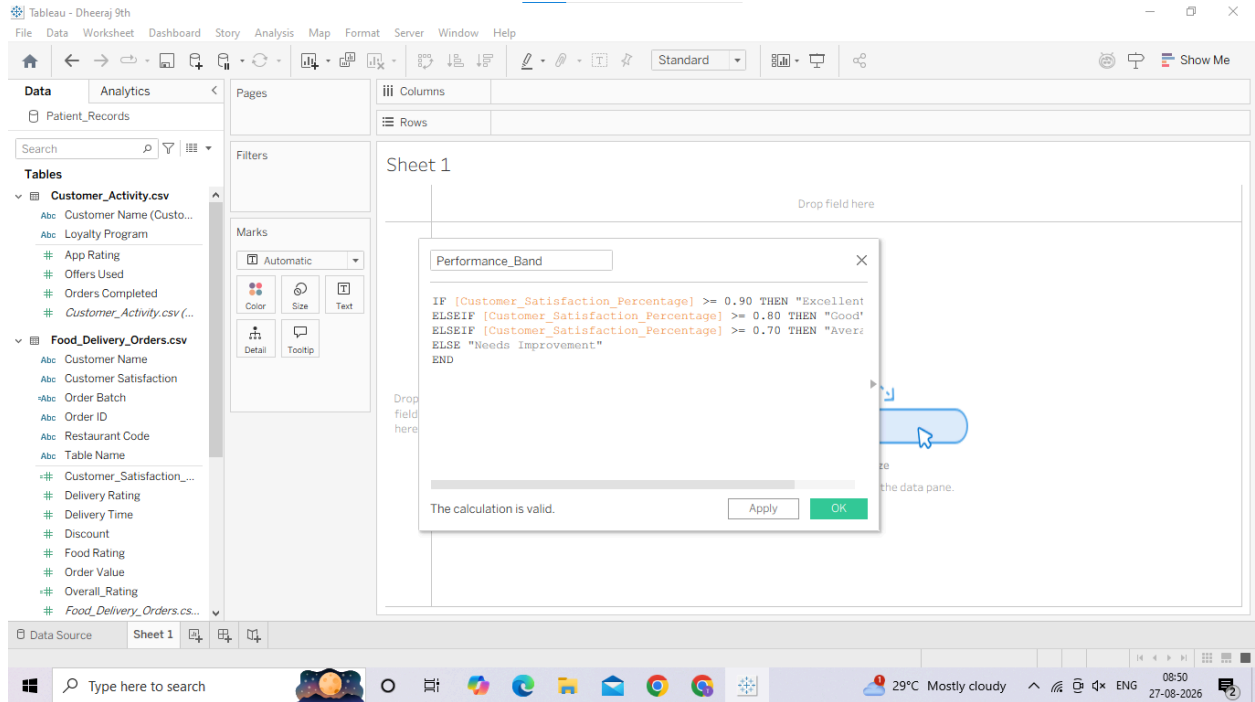
SHETH L.U.J. AND SIR M.V. COLLEGE

The image displays two screenshots of the Tableau Desktop interface, illustrating the process of formatting data in a worksheet.

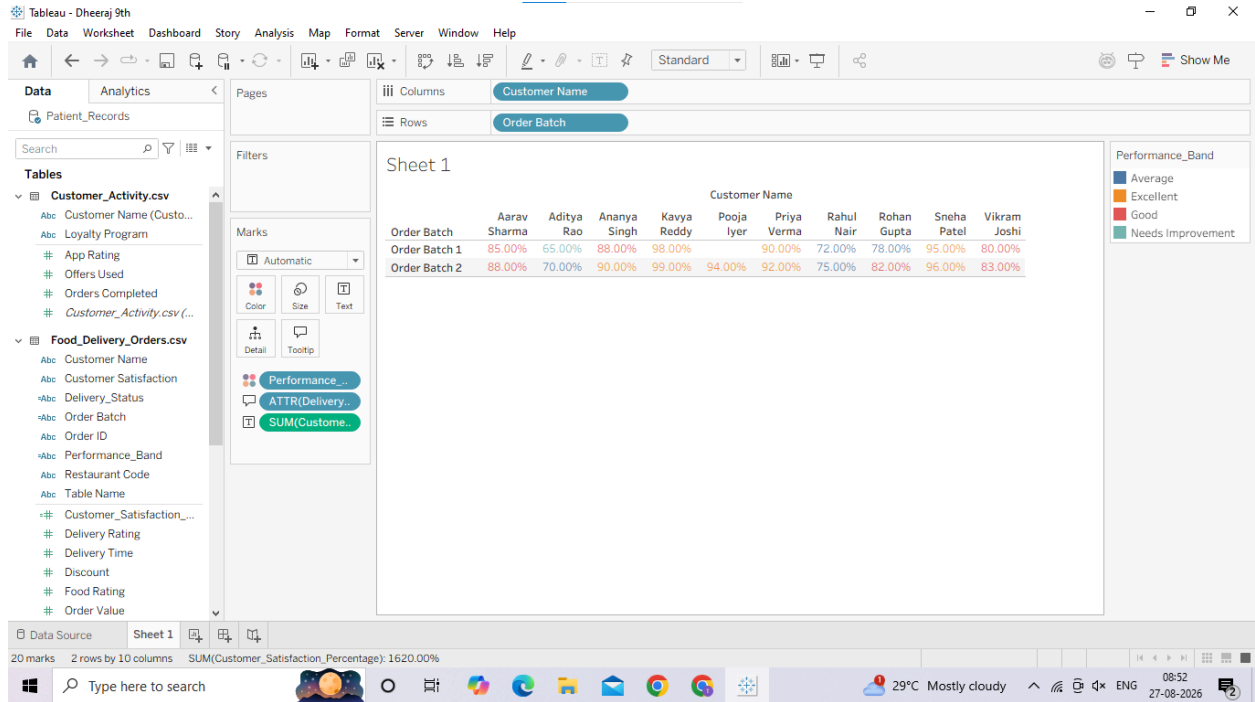
Top Screenshot: The Tableau Desktop window shows the 'Data' pane on the left with a list of tables. The 'Food_Delivery_Orders.csv' table is selected. The 'Add to Sheet' context menu is open, and the 'Number Format...' option is highlighted. The main view shows 'Sheet 1' with a 'Drop field here' prompt.

Bottom Screenshot: The 'Default Number Format [Multiple Fields]' dialog box is open. The 'Percentage' format is selected. The 'Decimal places' are set to 2. The 'OK' button is highlighted. The background shows the same Tableau Desktop window with 'Sheet 1' and the 'Drop field here' prompt.

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F: Handle Different Levels of Detail in Datasets

The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Customer Name' and the 'Rows' shelf contains 'Order Batch'. A calculation dialog box is open, showing the formula: `{ FIXED [Restaurant_Code] : AVG ([Customer_Satisfaction_Percentage]) }`. The dialog box has 'Apply' and 'OK' buttons. The background view shows a table with columns for 'Order Batch' and 'Customer Name' (Aarav Sharma, Aditya Rao, Ananya Singh, Kavya Reddy, Pooja Iyer, Priya Verma, Rahul Nair, Rohan Gupta, Sneha Patel, Vikram Joshi) and rows for 'Order Batch 1' and 'Order Batch 2'.

The screenshot shows the Tableau Desktop interface with the 'Default Properties' context menu open for the 'Restaurant_Average_Satisfaction' field. The menu options include: Add to Sheet, Show Filter, Cut, Copy, Edit..., Duplicate, Rename, Hide, Delete, Create, Convert to Discrete, Convert to Dimension, Change Data Type, Default Properties (selected), Geographic Role, Group by, Folders, Replace References..., and Describe... The 'Default Properties' sub-menu is also visible, showing options: Comment..., Color..., Number Format... (selected), Aggregation, and Total using. The background view shows a table with columns for 'Order Batch' and 'Customer Name' (Aarav Sharma, Aditya Rao, Ananya Singh, Kavya Reddy, Pooja Iyer, Priya Verma, Rahul Nair, Rohan Gupta, Sneha Patel, Vikram Joshi) and rows for 'Order Batch 1' and 'Order Batch 2'.

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Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home

Analytics

Pages

Columns Customer Name

Rows Order Batch

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Performance...

ATTR(Delivery...)

SUM(Custome...)

Tables

Loyalty Program

App Rating

Offers Used

Orders Completed

Customer_Activity.csv (...)

Food_Delivery_Orders.csv

Customer Name

Customer Satisfaction

Delivery_Status

Order Batch

Order ID

Performance_Band

Restaurant Code

Table Name

Customer_Satisfaction...

Delivery Rating

Delivery Time

Discount

Food Rating

Order Value

Overall_Rating

Restaurant_Average_Sat...

Sheet 1

Default Number Format [Restaurant_Average_Satisfaction_LOD]

Automatic

Number (Standard)

Number (Custom)

Currency (Standard)

Currency (Custom)

Scientific

Percentage

Custom

Percentage

Decimal places:

1

Clear

OK

Cancel

Performance_Band

Average

Excellent

Good

Needs Improvement

Rahul Nair

Rohan Gupta

Sneha Patel

Vikram Joshi

0.00%

78.00%

95.00%

80.00%

0.00%

82.00%

96.00%

83.00%

Data Source

Sheet 1

20 marks

2 rows by 10 columns

SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy

08:53

27-08-2026

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home

Analytics

Pages

Columns Customer Name

Rows Order Batch

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Performance...

ATTR(Delivery...)

SUM(Custome...)

Tables

Loyalty Program

App Rating

Offers Used

Orders Completed

Customer_Activity.csv (...)

Food_Delivery_Orders.csv

Customer Name

Customer Satisfaction

Delivery_Status

Order Batch

Order ID

Performance_Band

Restaurant Code

Table Name

Customer_Satisfaction...

Delivery Rating

Delivery Time

Discount

Food Rating

Order Value

Overall_Rating

Restaurant_Average_Sat...

Sheet 1

Customer Name

Order Batch

Aarav Sharma

Aditya Rao

Ananya Singh

Kavya Reddy

Pooja Iyer

Priya Verma

Rahul Nair

Rohan Gupta

Sneha Patel

Vikram Joshi

0.00%

95.00%

80.00%

0.00%

96.00%

83.00%

Customer_Overall_Average

FIXED [Customer_Name] : AVG([Customer_Satisfaction_Percentage])

The calculation is valid.

Apply

OK

Data Source

Sheet 1

20 marks

2 rows by 10 columns

SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy

08:54

27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Analytics

Search

Tables

- Abc Loyalty Program
- Abc App Rating
- Abc Offers Used
- Abc Orders Completed
- Abc Customer_Activity.csv (...)
- Food_Delivery_Orders.csv
- Abc Customer Name
- Abc Customer Satisfaction
- Abc Delivery_Status
- Abc Order Batch
- Abc Order ID
- Abc Performance_Band
- Abc Restaurant Code
- Abc Table Name
- Abc Customer_Overall_Average
- Abc Customer_Satisfaction_...
- Abc Delivery_Rating
- Abc Delivery_Time
- Abc Discount
- Abc Food_Rating
- Abc Order_Value
- Abc Overall_Rating

Columns: Customer Name

Rows: Order Batch

Sheet 1

Order Batch	Aarav Sharma	Aditya Rao	Ananya Singh	Kavya Reddy	Pooja Iyer	Priya Verma	Rahul Nair	Rohan Gupta	Sneha Patel	Vikram Joshi
Order Batch 1	85.00%	65.00%	88.00%	98.00%	90.00%	72.00%	78.00%	95.00%	80.00%	
Order Batch 2	88.00%	70.00%	90.00%	99.00%	94.00%	92.00%	75.00%	82.00%	96.00%	83.00%

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

20 marks 2 rows by 10 columns SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 08:55 27-08-2026

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Analytics

Search

Tables

- Abc Loyalty Program
- Abc App Rating
- Abc Offers Used
- Abc Orders Completed
- Abc Customer_Activity.csv (...)
- Food_Delivery_Orders.csv
- Abc Customer Name
- Abc Customer Satisfaction
- Abc Delivery_Status
- Abc Order Batch
- Abc Order ID
- Abc Performance_Band
- Abc Restaurant Code
- Abc Table Name
- Abc Customer_Overall_Average
- Abc Customer_Satisfaction_...
- Abc Delivery_Rating
- Abc Delivery_Time
- Abc Discount
- Abc Food_Rating
- Abc Order_Value
- Abc Overall_Rating

Columns: Customer Name

Rows: Order Batch

Sheet 1

Default Number Format [Customer_Overall_Average]

Automatic

- Number (Standard)
- Number (Custom)
- Currency (Standard)
- Currency (Custom)
- Scientific
- Percentage
- Custom

Percentage

Decimal places: 1

Clear OK Cancel

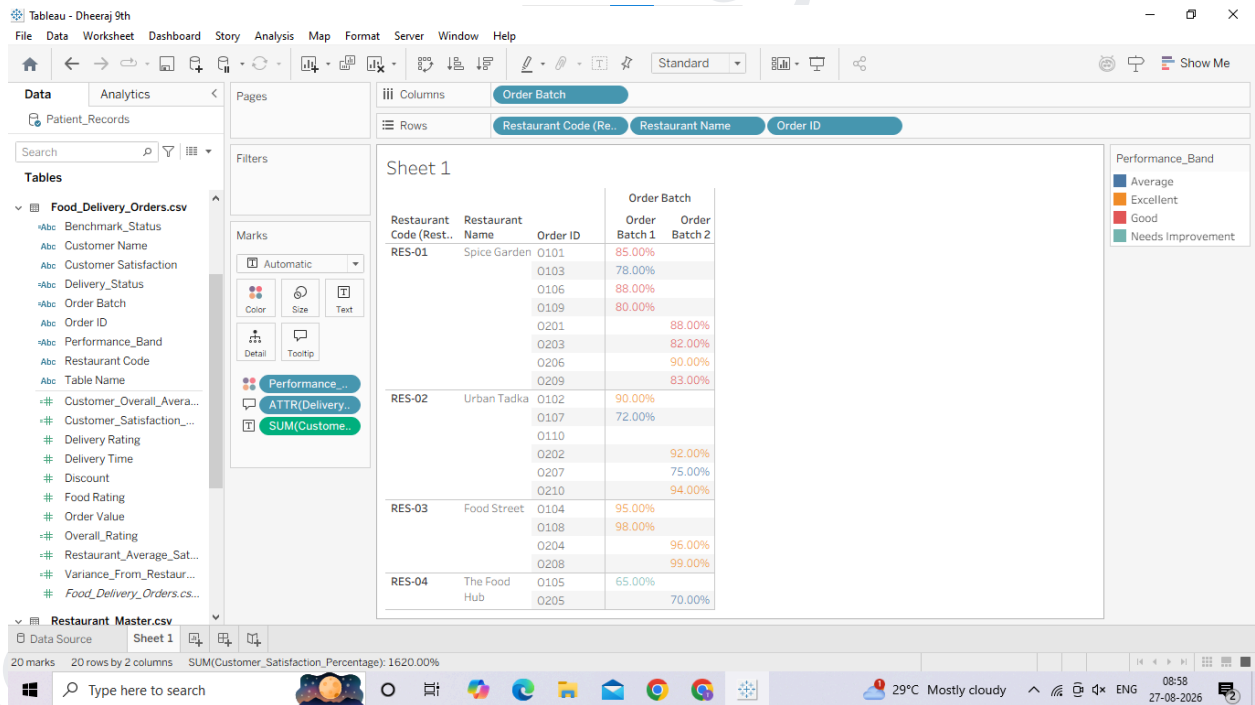
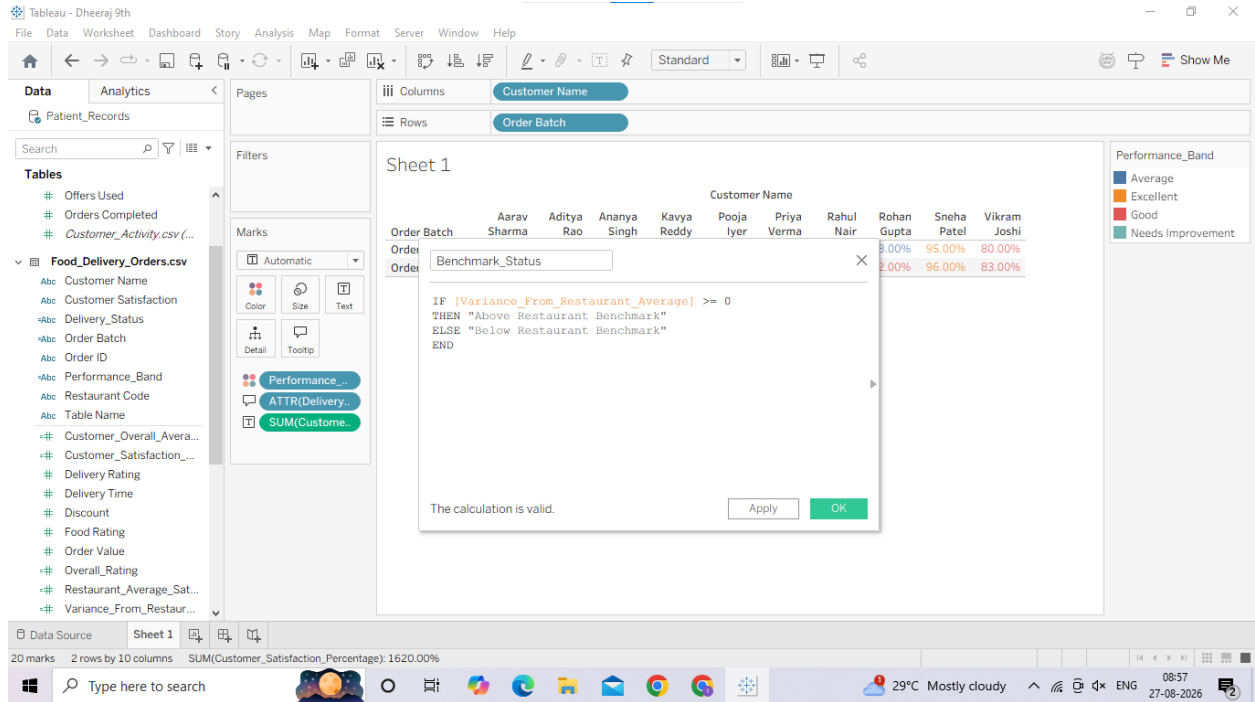
20 marks 2 rows by 10 columns SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 08:55 27-08-2026



SHETH L.U.J. AND SIR M.V. COLLEGE



G: Apply Advanced Data Cleaning Techniques

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

Pages

Columns Order Batch

Rows Restaurant Code (Rest... Restaurant Name Order ID

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Performance...

ATTR(Delivery...

SUM(Custome...

Performance_Band

Average

Excellent

Good

Needs Improvement

Sheet 1

Restaurant Restaurant Order Order
Code (Rest... Name ID Batch Batch 2

RES-0 Customer_Name_Clean

RES-0

RES-0

RES-0

The calculation is valid.

Apply OK

Restaurant Code (Rest...)	Restaurant Name	Order ID	Order Batch 1	Order Batch 2
0204			96.00%	
0208			99.00%	
RES-04	The Food Hub	0105	65.00%	
		0205	70.00%	

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction, Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 09:00 27-08-2026

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

Pages

Columns Order Batch

Rows Restaurant Code (Rest... Restaurant Name Order ID

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Performance...

ATTR(Delivery...

SUM(Custome...

Performance_Band

Average

Excellent

Good

Needs Improvement

Sheet 1

Restaurant Restaurant Order Order
Code (Rest... Name ID Batch Batch 2

RES-0 First_Name

RES-0

RES-0

The calculation is valid.

Apply OK

Restaurant Code (Rest...)	Restaurant Name	Order ID	Order Batch 1	Order Batch 2
0204			96.00%	
0208			99.00%	
RES-04	The Food Hub	0105	65.00%	
		0205	70.00%	

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction, Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 09:01 27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home

Analytics

Patent_Records

Search

Tables

- Food_Delivery_Orders.csv
 - Benchmark_Status
 - Customer Name
 - Customer Satisfaction
 - Customer_Name_Clean
 - Delivery_Status
 - First_Name
 - Order Batch
 - Order ID
 - Performance_Band
 - Restaurant Code
 - Table Name
 - Customer_Overall_Avera...
 - Customer_Satisfaction_...
 - Delivery Rating
 - Delivery Time
 - Discount
 - Food Rating
 - Order Value
 - Overall_Rating
 - Restaurant_Average_Sat...
 - Variance_From_Restaur...

Columns

Order Batch

Rows

Restaurant Code (Rest... Restaurant Name Order ID

Sheet 1

Restaurant Code (Rest... Restaurant Name Order ID Order Batch 1 Order Batch 2

RES-04 The Food Hub 0105 65.00%

0204 96.00%

0208 99.00%

0205 70.00%

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

TRIM(SPLIT([Customer_Name], " ", -1))

The calculation is valid.

Apply OK

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 09:01 27-08-2026

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home

Analytics

Patent_Records

Search

Tables

- Food_Delivery_Orders.csv
 - Benchmark_Status
 - Customer Name
 - Customer Satisfaction
 - Customer_Name_Clean
 - Delivery_Status
 - First_Name
 - Last_Name
 - Order Batch
 - Order ID
 - Performance_Band
 - Restaurant Code
 - Table Name
 - Customer_Overall_Avera...
 - Customer_Satisfaction_...
 - Delivery Rating
 - Delivery Time
 - Discount
 - Food Rating
 - Order Value
 - Overall_Rating
 - Restaurant_Average_Sat...
 - Variance_From_Restaur...

Columns

Order Batch

Rows

Restaurant Code (Rest... Restaurant Name Order ID

Sheet 1

Restaurant Code (Rest... Restaurant Name Order ID Order Batch 1 Order Batch 2

RES-04 The Food Hub 0105 65.00%

0204 96.00%

0208 99.00%

0205 70.00%

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

SPLIT([Restaurant_Code], "-", 1)

The calculation is valid.

Apply OK

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 09:02 27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Back Forward Refresh Save Undo Redo Copy Paste Paste as Table Standard Show Me

Data Analytics Pages Columns Order Batch Rows Restaurant Code (Re.. Restaurant Name Order ID

Search Patient_Records

Tables

- Food_Delivery_Orders.csv
 - Benchmark_Status
 - Customer Name
 - Customer Satisfaction
 - Customer_Name_Clean
 - Delivery_Status
 - First_Name
 - Last_Name
 - Order Batch
 - Order ID
 - Performance_Band
 - Restaurant Code
 - Restaurant_Prefix
 - Table Name
 - Customer_Overall_Avera...
 - Customer_Satisfaction_...
 - Delivery_Rating
 - Delivery Time
 - Discount
 - Food Rating
 - Order Value
 - Overall_Rating
 - Restaurant_Average Sat...

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Performance... ATTR(Delivery... SUM(Custome...

Sheet 1

Restaurant Restaurant Order Batch
Code (Rest.. Name Order ID Order Batch 1 Order Batch 2

RES-0 Restaurant_ID

SPLIT([Restaurant_Code], "-", 2)

RES-0

RES-0 The calculation is valid. Apply OK

RES-04 The Food Hub O204 96.00%
O208 99.00%
O105 65.00%
O205 70.00%

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

Data Source Sheet 1

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 09:02 27-08-2026

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Back Forward Refresh Save Undo Redo Copy Paste Paste as Table Standard Show Me

Data Analytics Pages Columns Order Batch Rows Restaurant Code (Re.. Restaurant Name Order ID

Search Patient_Records

Tables

- Food_Delivery_Orders.csv
 - Benchmark_Status
 - Customer Name
 - Customer Satisfaction
 - Customer_Name_Clean
 - Delivery_Status
 - First_Name
 - Last_Name
 - Order Batch
 - Order ID
 - Performance_Band
 - Restaurant Code
 - Restaurant_ID
 - Restaurant_Prefix
 - Table Name
 - Customer_Overall_Avera...
 - Customer_Satisfaction_...
 - Delivery_Rating
 - Delivery Time
 - Discount
 - Food Rating
 - Order Value
 - Overall_Rating
 - Restaurant_Average Sat...

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Performance... ATTR(Delivery... SUM(Custome...

Sheet 1

Restaurant Restaurant Order Batch
Code (Rest.. Name Order ID Order Batch 1 Order Batch 2

RES-0 Order_Display_Label

[Order_ID] + "-" + [Customer_Name_Clean] + " (" + [Restaur...

RES-0

RES-0 The calculation is valid. Apply OK

RES-04 The Food Hub O204 96.00%
O208 99.00%
O105 65.00%
O205 70.00%

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

Data Source Sheet 1

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction_Percentage): 1620.00%

Type here to search

28°C Mostly cloudy 09:03 27-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

